

Business Document Ingestion — Technical Documentation

Architecture, Methodology, Configuration, and Operating Runbook

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Technical Documentation

Purpose and audience

This is the operator and maintainer manual for the Business Document Ingestion repository. A new operator should be able to use it without prior chat history. It explains the system boundary, why each gate exists, how a controlled run moves from source PDFs to approved facts, how to recover from failure, and where every tunable or artifact contract is defined.

Version 1.0.0 is the stable evidence-control baseline and vendor-neutral post-review data boundary. This manual also documents backward-compatible work listed under **Unreleased** in `CHANGELOG.md`; this checkout may include unmerged development capabilities as identified in `HANDOFF.md`. They are not part of the immutable 1.0.0 release record until a later stable version is prepared and tagged. Neither scope is a configured client production service. Live provider selection, privacy and retention approval, representative golden-set measurement, target CRM delivery, and public tenant-isolated retrieval remain client-specific work.

Courtesy credit is recorded in `NOTICE` for Christopher Melhauser and theonlymuffinbot. The latter describes collaborative AI work produced through a mix of Anthropic Claude Opus 5, OpenAI GPT-5.6 Terra and Sol, and xAI Grok 4.5 models. It is descriptive tooling context, not AI legal authorship or ownership.

How to use the documentation

This guide owns the end-to-end explanation. More detailed contracts have one authoritative home:

Question	Authority
What does each environment setting do?	<code>references/runtime-configuration.md</code>
Which commands and invocation controls exist?	<code>references/command-line-reference.md</code> , the complete parser-derived <code>references/cli-help-catalogue.md</code> , and each command's <code>--help</code>
What exact files and fields must a stage read or write?	<code>references/artifact-contracts.md</code>
What must be true before a phase may advance?	<code>references/workflow-gates.md</code>
How are records and derived fields modeled?	<code>references/data-model.md</code> , <code>references/extraction-schema.md</code> , and <code>references/derived-field-schema.md</code>
How are common CRM import files mapped and safely written?	<code>references/crm-write-readiness.md</code>
How is the approved-fact MCP/API set up, deployed, accepted, and used?	<code>skills/mcp-api-operations/SKILL.md</code> and <code>references/mcp-production-integration.md</code>

Question	Authority
How do image proposals and source-only handoff work?	<code>skills/record-intake/SKILL.md</code> and <code>references/visual-ingestion.md</code>
Which record-write and advanced-analytics features are still planned?	<code>references/business-data-platform-roadmap.md</code>
How does a specialized lane operate?	Its phase file in <code>references/</code>
What is the current verified state?	<code>HANDOFF.md</code> ; <code>docs/REPOSITORY_AUDIT.md</code> is a dated historical audit record

If code, executable help, and prose disagree, stop the run. Preserve all output, record the inconsistency, and correct the code, tests, and authority document together.

Examples that use `python` assume the project virtual environment is active. Run `. .venv/bin/activate`, or substitute `.venv/bin/python`.

System boundary

Non-negotiable controls

1. Preserve every source page and original extracted value.
2. Store corrections as evidence-linked amendments; never replace evidence.
3. Emit a record or explicit exception for every input page and failed stage.
4. Require independent evidence for consensus; repeated prompts to one model are correlated readings, not independent voters. Independence is a property of the model vendor, not the transport, so a router serving another vendor's model counts as that vendor.
5. Treat provider confidence as provenance only. It cannot approve a fact and is never a term in a decision.
6. Prove applicable arithmetic and report attribution and completeness separately. No one result implies another.
7. Keep protected findings in final review. A grouping, dependency, threshold, or batch decision cannot clear them.
8. Admit only approved, provenance-linked facts with no open review to canonical retrieval and downstream staging.
9. A control that processed nothing has not passed. Empty or unusable input produces an explicit exception and a refusal, never a clean result.
10. A value outside its plausible range is a misread figure, not a policy question. Route it to review, and never infer a unit, a date order, or a country from magnitude alone.
11. Transmission authorization applies only to its named run, source set, and configured providers. It never bypasses a phase prerequisite, permits a skipped item, accepts a proposal, or authorizes production actions.

Evidence and approval layers

The controlled progression is:

immutable source → **raw observations** → **normalized proposals** → **deterministic controls** → **exceptions and decisions** → **final gate** → **approved canonical facts** → **no-send staging and read-only retrieval**

A failed final gate returns work to exceptions and decisions; it never jumps to canonical facts.

The layers are intentionally rebuildable:

- **Source** contains byte-retained originals and one-page masters.
- **Raw** contains provider responses, native text, image variants, and other direct observations. These are evidence, not facts.

- **Proposal** contains typed readings, mappings, groupings, and amendments with source references.
- **Control** contains consensus, arithmetic, validation, entity, attribution, completeness, and sampling results.
- **Review** contains every unresolved item and any client or LLM proposal.
- **Canonical** contains only authorized, reconciled, review-clear facts.

Any later layer can be rebuilt from retained earlier evidence. A rerun must use a new output path and must not erase the prior attempt.

Implemented lanes and deliberate limits

Lane	Implemented behavior	Boundary
Intake	Source hashing, conservative page splitting, ordered provenance, native-text retention, high-signal classification	Uncertain pages remain queued; classification is not production party-role detection
Optional visual source intake	Seven schema/image/session/proposal operations over MCP and OAuth JSON API; append-only private journal; local verified source export	Pre-pipeline proposals only; no approval, canonical write, native capture client or automatic extraction handoff
Image preparation	Grayscale master plus sibling enhanced/binarized variants	Never replaces the page master; JBIG2 numerics remain suspect
LLM extraction	Configured LLM provider; strict schema, raw retention, retries, throttling, cache	One provider per invocation; proposal voter only
Table comprehension	Source profile, rows, audit, corpus layout context, bounded rereads	Same-model roles are not consensus; rereads are amendments
Document AI	OCR tokens, coordinates, tables, raw page response	Corroborating evidence only; never maps or approves protected facts
Google handwriting OCR	Page-complete Cloud Vision OCR, retained renders/raw responses, separately detected region binding	One proposal-only HTR provider-group vote; cannot detect/validate itself or overwrite print
Reconciliation	Independent consensus, exact-registry table comparison, arithmetic, deterministic validation	Provider/schema failures and empty records fail closed
Handwriting	Independent provider-group HTR-result reconciliation	Financial readings remain review-required amendments; same-provider roles count once
Business controls	Entity resolution, configurable attribution, completeness, sampling, descriptive analytics	Missing/ambiguous reconciliation blocks; analytics do not approve facts

Lane	Implemented behavior	Boundary
Page verification	Stratified page sample; each page's image checked against the rows the export claims for it, with lost and invented money named mechanically; corrections graded by the independent evidence beside them and applied as append-only amendments	A correction that would break a line's own base x rate = commission is refused, and one the reviewer alone supports cannot be applied
CRM input	Every field exported once with the status that governs it; empty cells filled beside the reading from the run's own evidence and cited public sources; validation against the canonical schema	Not canonical; an inference never overwrites a reading, and a validation finding is a mapping decision, not a repair
Dealer locations	Optional Google Places lookup of dealer and brand names, one request per party within a cap, every candidate retained	Business names only; a candidate is accepted only when it accounts for every word of the name, and lands beside the reading as a probable public fact
Review reduction	Card reviewer, cross-record search, dataset agent, grouping, safe consolidation, reproducible client package	Presentation and proposal reduction only; source queue remains retained
Post-return proposals	Inferred-control artifacts, preserved client-response context, bounded reference discovery, Primary-LLM/independent-Secondary-LLM iterations, and deterministic evidence-graph queries	Never independent consensus, authoritative GL/payment evidence, gate clearance, or fact approval
Post-review data	PostgreSQL plan, load plan, atomically verified no-send target package, shared SQLite/MCP/HTTPS CRM query and report layer	Only approved facts; no target CRM receiver or public deployment is implied

Installation and repository verification

Optional source-image intake and handoff

This is a pre-pipeline boundary, not a new independent extraction lane. The connected vision-capable assistant reads retained PNG/JPEG pages and submits source-cited candidates against `get_ingestion_schema`. The server makes no model-provider call and needs no additional model API key. The host model's self-reported identity is provenance, not an authenticated independent vote.

The existing servers still require an approved non-empty retrieval snapshot, even for an intake-only caller. Satisfy the Phase 6 activation gate for that snapshot and obtain separate intake/source-model data-flow authorization. Do not construct an empty or unfinished snapshot to bypass activation.

Surfaces and activation parameters

Surface	Baseline	Intake opt-in
Local stdio MCP	12 approved-fact tools; shell launcher	Direct <code>retrieval_mcp.py</code> with <code>--ingestion-dir</code> ; 23 tools
Remote OAuth MCP	14 tools including export jobs	<code>retrieval_remote_mcp.py --ingestion-dir</code> ; up to 25 tools, scope-filtered
TLS/bearer REST	Read-only GET routes	Unchanged; no intake support
Remote OAuth JSON API	Intake routes absent	Seven <code>POST /api/ingestion/{operation}</code> routes
Internal FastAPI sidecar	Private localhost or same-task retrieval/report API	<code>retrieval_sidecar.py</code> ; no intake support

`--ingestion-dir` is a private journal directory separate from the repository, approved snapshot and run. Existing journal/root permissions must be private; new directories/files use 0700/0600. Local `--ingestion-tenant` defaults to `local`, and local ownership is fixed to `local-operator`. Remote journal identity comes from the required `--tenant`; session ownership comes from the validated OAuth `sub`, never caller arguments. An intake-only identity needs the applicable `ingestion:read/ingestion:submit` scope in addition to all ordinary audience, role and tenant checks. Neither grants CRM access.

Remote `--max-request-bytes` defaults to 1,000,000 for the entire JSON/base64 envelope. Set at least 14,000,000, with matching proxy/client limits, to transfer a maximum 10,000,000-byte original image. The shared default rate limit remains 60 requests per rolling minute per client address. Intake adds no environment setting. Full server arguments remain in executable help and the parser-derived CLI catalogue; all intake-specific flags and payload fields are defined in **Visual Intake**.

`get_crm_capabilities` describes the immutable approved-fact service, not overall intake enablement. Remote `/health` and `/docs` show `read_only: false` when intake is enabled and retain `canonical_read_only: true`.

Operations, outputs and limits

Read operations are `get_ingestion_schema`, `list_ingestion_sessions`, `get_ingestion_page`, `get_ingestion_status`, `list_ingestion_session_reviewers`, `get_ingestion_review_summary`, and `get_record_proposal`. Submission operations are `create_ingestion_session`, `upload_ingestion_page`, `submit_record_proposal`, and `grant_ingestion_session_reviewer`. The JSON API takes the argument object directly; MCP uses `tools/call`. MCP image retrieval returns `ImageContent` plus metadata; the application API returns base64 in `data`. Clients must transfer real bytes, not ask a language model to construct base64. Native attachment relay, cross-origin browser upload UI, owner transfer, and admin moderation are not shipped. A thin same-origin review page is now served at `/ingestion/review`; it uses the existing intake JSON API with an operator-supplied bearer token rather than a separate browser auth layer.

The `visual_ingestion_v1` schema reuses controlled extraction header/line fields and document types. Each observation retains value, raw text, printed/handwritten source, page number and normalized box in encoded-image orientation. Preserve unknown labels as `unmapped_fields`, unresolved readings as null plus issues, and unreadable pages as explicit exceptions. Every declared page is accounted for; zero extracted records cannot validate successfully. Approval/control fields are forbidden. Structural validation proves neither reading accuracy nor independent consensus. All outcomes remain unapproved and unpublished.

For a co-located internal application, `retrieval_sidecar.py` provides a thin FastAPI wrapper over `crm_service.py`. It binds one immutable approved-fact SQLite snapshot resolved from `RETRIEVAL_DB_URL`, `RETRIEVAL_ARTIFACT_ROOT`, or an explicit `--database` argument, and is intended for localhost or same-task container use rather than public internet exposure. Its contract is `GET /health`, `GET /capabilities`, and `POST` JSON routes for account cards, record search/query, sales analysis, and standard reports.

Limits are 1-20 pages/session; single-frame PNG/JPEG only; 10 MB original bytes and 25 million pixels/image; 1 MB canonical JSON/proposal; 100 records/proposal; 200 lines/record; 100 unmapped fields and 100 issues/record; 4,000 characters per observation value/raw reading and 1,000 per issue/page-exception reason. The journal allows 10,000 entries and 250 MB accounted payload/result/image bytes, with a conservative 1 MB result reservation per insert; a session allows 100 page/proposal entries including rejections. These fixed decimal-byte bounds are not tuning flags or physical disk-size guarantees.

The journal retains bounded corrupt/unsupported image attempts and rejected proposals. Malformed/oversized requests and exhausted quotas are refused before acceptance; the sender retains originals and refusal receipts. Exact-key/input replay returns the original receipt across restart and API/MCP. Changed content needs a new key. A corrected proposal appends; replacing an accepted image slot requires a new session. A later proposal never clears an earlier finding. HTTP 200 can carry a retained rejection: inspect **status** and **findings**.

Source export and pipeline re-entry

The local `visual_ingestion_export.py` utility opens an existing journal read-only and snapshots one owner/session. `export` requires positional journal directory, `--tenant`, `--owner`, `--session-id` and new `--out-dir`; no defaults or overwrite/resume are supplied. Local OS permissions grant operator access; `--owner` selects the session and is not remote authentication. Invoke export through a fresh `run_workspace.py` root and use absolute script/interpreter paths. Journal and destination must not overlap.

The `visual_intake_source_package_v1` output preserves every upload's original bytes in `originals/*.bin`, all payloads/receipts/proposal versions in `journal.json`, final-review `exceptions.json`, and a completion `manifest.json` binding every path, size, checksum, count and transformation. A complete usable source set additionally produces `visual_<session-identity-hash>.pdf`; its page map matches the real intake producer's stable IDs. Never rename that PDF.

PDF preparation keeps encoded orientation and page order, does no EXIF rotation, crop, resize or OCR, converts display pixels to RGB with transparency on white, and applies no ICC correction. Original bytes remain authoritative. Its 300-DPI layout is not evidence of physical dimensions or original scan resolution. Preparation is capped at 50 million total pixels. Inspect all pages visually.

Keep the export receipt separately. `verify PACKAGE --manifest-sha256 HASH` requires that receipt's exact manifest SHA-256, not a newly computed untrusted hash. It checks inventory, safe paths, symlinks, file/entry integrity and size bounds (1 MB manifest; 250 MB/file). Export and verify exit 0 only for `source_prepared_review_required`; exit 2 retains blocked source preparation, including an otherwise byte-intact archive; exit 1 indicates refusal/error. Incomplete directories and failed partial PDFs remain preserved, never resumed or used. Retry into a new directory. Hashes are not privileged-actor signatures.

Only a verified package with a declared non-null `source_pdf` proceeds to `scan_profile.py`, then `ingest_pages.py`, under the run wrapper. Bind the full source package and derivative in operations provenance and include its `exceptions.json` with every later review-bearing artifact. Continue the normal ordered extraction and control sequence. No host-model proposal is promoted to a producer-authenticated extraction handoff or independent consensus vote.

Repository intake tests and the source-handoff tests supplement the six-gate disabled-intake acceptance. Deployment additionally needs real TLS/OAuth/client upload, image reading, quota/retry/restart, backup/restore and revocation tests. Record writes, publication, native capture and expanded analytics remain explicit roadmap work, not capabilities proved by code coverage.

Dependencies

- Python 3.12 or newer.
- Development packages from `requirements-dev.txt`.
- Poppler tools (`pdftotext`, `pdftopdf`) for acceptance and visual verification.
- Tectonic 0.17.0 for deterministic tracked PDF generation.

On macOS, brew install poppler tectonic supplies the document tools. In other environments, use an approved package source and verify the selected Tectonic version before release generation.

Full quality gate

Use the wrapper in a normal terminal or CI worker. It deliberately has no interactive-output deadline; the workflow's 45-minute CI job timeout is the separate safety boundary. A tool that stops displaying output does not make the underlying command incomplete—wait for its process exit status or run this wrapper directly:

```
PYTHON_BIN=.venv/bin/python bash scripts/quality_gate.sh
```

The expanded sequence below prepares the environment, then reproduces the wrapper's checks. The suite runs once under branch coverage; a separate bare pytest pass is not part of the wrapper.

```
python3 -m venv .venv
.venv/bin/python -m pip install -r requirements-dev.txt
.venv/bin/python -m pip check
quality_tmp_dir="$(mktemp -d)"
COVERAGE_FILE="$quality_tmp_dir/.coverage" PYTEST_ADDOPTS="-p no:cacheprovider" \
.venv/bin/python -m coverage run --branch -m pytest
COVERAGE_FILE="$quality_tmp_dir/.coverage" \
.venv/bin/python -m coverage report --fail-under=100
RUFF_CACHE_DIR="$quality_tmp_dir/ruff-cache" .venv/bin/ruff check .
RUFF_CACHE_DIR="$quality_tmp_dir/ruff-cache" \
.venv/bin/ruff format --check scripts tests
.venv/bin/python scripts/release_check.py
rm -rf "$quality_tmp_dir"
```

The required threshold is 100% statement and branch coverage for `scripts/`. The release check also verifies required documents, local links, tracked Markdown/LaTeX/PDF synchronization, repository hygiene, complete environment documentation, and the complete argparse command index.

Configuration

Precedence and reproducibility

Copy `.env.example` to the Git-ignored root `.env`. Effective settings resolve in this order:

1. Explicit command-line option.
2. Process environment, CI variable, or secret-manager injection.
3. Repository-root `.env`, unless the process exports `BUSINESS_DOCUMENT_IGNORE_PROJECT_ENV` as true.
4. Safe code default.

Use `operations.py manifest --config-from-env` for a live run. It records the effective non-secret configuration visible to the process. Secret values are redacted; credential-variable names are retained. Paths and client inputs stay on the command line because they are run evidence, not shared configuration.

`references/runtime-configuration.md` documents every tracked environment setting individually: default, range or allowed values, effect, security boundary, and change guidance. It also documents the process-only `PYTHON_BIN`, `BUSINESS_DOCUMENT_RETRIEVAL_DB` and `BUSINESS_DOCUMENT_IGNORE_PROJECT_ENV` variables, which are deliberately not stored in `.env`; the last switches the root `.env` off for a process and the children it starts. `references/command-line-reference.md` indexes every CLI and subcommand. Its companion `references/cli-help-catalogue.md` is the complete flag-level technical appendix: it reproduces every command and subcommand's positional inputs, options, defaults, choices, and help text from the live parser. Only `usage`: line wrapping is normalized across supported CPython versions; the executable's `--help` remains the exact runtime display. The release gate fails if the catalogue drifts. The exact parser help is available with:

```
python scripts/COMMAND.py --help
python scripts/COMMAND.py SUBCOMMAND --help
```

Provider and lane selection

The run-level registry separates these roles:

- `LLM_EXTRACT_PROVIDER` selects the ordinary extraction provider.
- `LLM_CONSENSUS_PRI_PROVIDER` and `LLM_CONSENSUS_SEC_PROVIDER` select two separately invoked, genuinely independent consensus lanes.
- `LLM_REASONING_PROVIDER` selects proposal-only reasoning lanes.
- `LLM_CLIENT_REVIEW_PROVIDER` may override the card reviewer.
- `LLM_POST_REVIEW_PROVIDER` may override both post-review reasoning lanes.
- `AI_SIMULATED_CLIENT_REVIEW_PROVIDER` selects the separate simulated-client draft-comment lane; its selected model and all retry/packet controls use the corresponding `AI_SIMULATED_CLIENT_REVIEW_*` settings.

Each provider resolves its model from the corresponding extraction, consensus, or reasoning model setting. The final reviewer can take invocation-only `--final-provider` and `--final-model` overrides, but all context, timeout, retry, reasoning, and safety controls remain inherited. A model change requires schema/capability verification and representative golden-set evaluation.

Shared throttling uses provider request/token ceilings, a safety ratio, an inter-request interval, and preflight request-size bounds. Concurrency changes throughput, not approval. An oversized packet fails before network I/O. A terminal reviewer rate limit opens a circuit breaker so remaining cards are retained without further calls.

Credentials

Do not store a provider response, token, API key, service-account JSON value, or credential file in `.env`, a run directory, a handoff, or version control. `.env` may contain approved API-key values for local execution, but it is not a secret manager and must remain private. Prefer external injection.

Before a Google-backed run:

```
python scripts/reauthorize_google.py
```

The helper refreshes gcloud user credentials and Application Default Credentials, verifies both token paths, optionally offers to enable required APIs, and can write short-lived Document AI and Cloud Vision token variables to a user-selected 0600 file outside the repository. If `GOOGLE_APPLICATION_CREDENTIALS` is set, verify that it points to the intended external file; otherwise Vertex uses refreshed ADC.

Run layout and artifact discipline

Use a dedicated directory per run. Initialize it with `python scripts/run_workspace.py init RUN_DIR`; then run every writing command through `python scripts/run_workspace.py run RUN_DIR -- ...`. The wrapper makes `RUN_DIR/...` (and the operations guides' `RUN/...`) workspace-relative, rejects declared output-path escapes, and pins LLM cache and throttle state to `RUN_DIR/runtime/`. Source inputs and credentials intentionally remain outside the run root as read-only boundaries. A practical layout is:

<code>RUN_DIR/</code>	
<code>pages/</code>	retained one-page masters and source copies
<code>providers/</code>	normalized records, handoffs, raw responses, and exceptions
<code>tables/</code>	source-native table profile and corpus artifacts
<code>htr/</code>	handwriting images, annotations, and decisions
<code>controls/</code>	consensus, arithmetic, validation, and reconciliation
<code>templates/</code>	observed source templates and drift artifacts
<code>graph/</code>	evidence-graph overlays

relationships/	retained relationship-proposal passes
review/	exhaustive queue and proposal-only review artifacts
client/	context, issued package, and preserved returned copy
analytics/	local analysis outputs
canonical/	approved export, load plan, staging, and retrieval DB
delivery/	reproducible delivery package artifacts
runtime/	per-run cache and throttle state
retries/	no-clobber recovery overlays
logs/	initialization, command ledger, and audit artifacts

Operational manifests, adapter contracts, privacy inventories, reviewer exports, client packages, deployment/load plans, and retrieval stores are no-clobber. Resumable stage state is the intentional exception: it updates atomically and remains bound to the same manifest hash. Exact filenames and schemas are in [references/artifact-contracts.md](#).

Before the first provider call for a full run, retain a non-secret `RUN_DIR/logs/run_authorization.md` with the run identifier, source identifiers/hashes, timestamp, selected lanes, provider families, and whether every in-scope source item may be sent. This records transmission scope only; it does not accept a proposal, clear a control, enable auto-accept/carry-forward, or authorize canonical, CRM, or delivery work. Execute phases as a strict state machine: complete and inspect upstream artifacts before starting a dependent lane, run only genuinely independent same-phase lanes in parallel, and retain every refusal, schema mismatch, provider failure, missing prerequisite, or zero-coverage result. At closure, run workspace audit and required lane coverage before building the final queue from all review-bearing base and recovery artifacts.

Use relative output paths inside the wrapper, for example:

```
python scripts/run_workspace.py run RUN_DIR -- \
  python /ABSOLUTE/REPOSITORY/PATH/scripts/scan_profile.py SOURCE \
  --out profile.json
```

Operating runbook

The examples below show control order. Replace placeholder paths with a new authorized run directory. Use each command's `--help` before execution.

This is an evidence and control flow, not a data-overwrite flow. Every optional LLM result remains a retained proposal until the normal authorization and downstream-control path completes.

1. Discover and authorize

Before receiving client data, complete `docs/CLIENT_USER_GUIDE.md`:

- source owner, transfer route, dedicated source area, and retention/deletion;
- corpus size, date range, document families, scan characteristics, and known duplicate behavior;
- business questions, attribution definition, GL/payment/reference exports;
- approved providers, regions, credentials, request/cost caps, and caching;
- golden-set design, acceptance thresholds, client reviewer, and target system;
- whether retrieval remains local or needs separately accepted remote controls.

Do not begin live provider work until privacy, residency, and provider approval are explicit.

2. Profile, retain, split, and prepare

```
python scripts/scan_profile.py SOURCE --out RUN_DIR/operations/scan_profile.json
python scripts/ingest_pages.py SOURCE.pdf --out RUN_DIR/intake
python scripts/reassemble_pages.py RUN_DIR/intake/ingestion_manifest.json \
  --out RUN_DIR/controls/reassembly.json \
  --exceptions RUN_DIR/review/reassembly_exceptions.json
python scripts/preprocess_pages.py RUN_DIR/intake/pages \
```

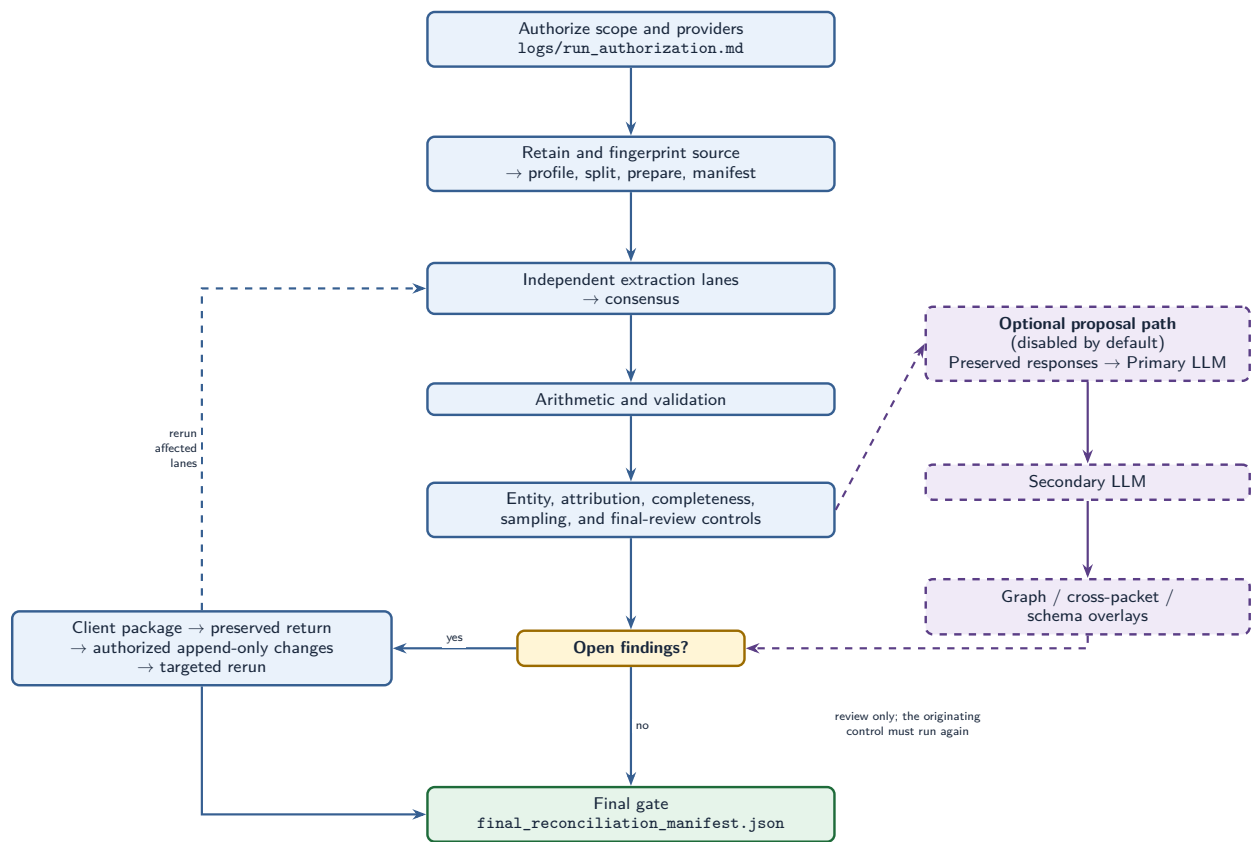


Figure 1: Operating runbook control order. This is an evidence and control flow, not a data-overwrite flow; the optional proposal path adds overlays and never clears the control that raised a finding.

```
--out RUN_DIR/preprocessing \
--manifest RUN_DIR/preprocessing/preprocessing_manifest.json
```

The source is never modified. Inspect page counts, hashes, relative paths, one-page boundaries, order, scan warnings, reassembly proposals, duplicate candidates, and all classification exceptions before extraction.

3. Create the run manifest and state

```
python scripts/operations.py manifest \
  RUN_DIR/intake/ingestion_manifest.json \
  --config-from-env --out RUN_DIR/operations/run_manifest.json
python scripts/operations.py stage RUN_DIR/operations/run_state.json profile \
  --manifest-hash MANIFEST_SHA256
python scripts/operations.py stage RUN_DIR/operations/run_state.json intake \
  --manifest-hash MANIFEST_SHA256
```

Keep the manifest hash constant for resumed stages. A changed input or effective configuration requires a new run.

4. Produce independent extraction evidence

Invoke the primary and secondary lanes separately:

```
python scripts/llm_adapter.py RUN_DIR/intake/ingestion_manifest.json \
  --lane consensus_primary \
  --out RUN_DIR/providers/primary_engine.json \
  --adapter-out RUN_DIR/providers/primary_handoff.json \
  --exceptions RUN_DIR/review/primary_exceptions.json \
  --raw-dir RUN_DIR/providers/primary_raw

python scripts/llm_adapter.py RUN_DIR/intake/ingestion_manifest.json \
  --lane consensus_secondary \
  --out RUN_DIR/providers/secondary_engine.json \
  --adapter-out RUN_DIR/providers/secondary_handoff.json \
  --exceptions RUN_DIR/review/secondary_exceptions.json \
  --raw-dir RUN_DIR/providers/secondary_raw
```

The two lane providers must be genuinely independent. Validate each adapter handoff and retain every raw response and exception. Two empty or failed engine records are never agreement.

For source-native tables, follow `references/table-comprehension.md`. Profile layout, propose source rows with cell evidence, run the non-independent audit, optionally invoke Document AI for a conditional buddy audit, then map only with an exact client-approved registry. Corpus refinement is sequential because source-layout context is order-dependent. Every changed reread is a linked amendment proposal.

5. Reconcile and validate

```
python scripts/consensus.py \
  RUN_DIR/providers/primary_engine.json \
  RUN_DIR/providers/secondary_engine.json \
  --out RUN_DIR/controls/consensus.json \
  --exceptions RUN_DIR/review/consensus_exceptions.json
python scripts/arithmetic_check.py RUN_DIR/controls/consensus.json \
  --out RUN_DIR/controls/proofed.json
python scripts/adjudicate.py RUN_DIR/controls/proofed.json \
  RUN_DIR/providers/primary_engine.json \
  RUN_DIR/providers/secondary_engine.json \
  --out RUN_DIR/review/amendment_proposals.json \
  --exceptions RUN_DIR/review/adjudication_exceptions.json
python scripts/validate_extraction.py RUN_DIR/controls/proofed.json \
```

```
--out RUN_DIR/controls/validated.json \
--exceptions RUN_DIR/review/validation_exceptions.json
```

Consensus accepts only `independent_extraction_handoff_v1` artifacts from distinct provider groups and consensus lanes. It verifies embedded record and retained raw-response hashes, compares typed values without coercing identifiers into numbers, rejects duplicate engine/document records, and carries a voted document type or `unknown`. Arithmetic compares source-visible equations within the configured absolute tolerance. Adjudication is bounded and amendment-only. Validation never silently normalizes a failed field into passing status.

Source-template observations

Three lanes read a source-template observation artifact and nothing else produces one: `schema_discovery.py` discover, `allocation_policy.py` discover, and `template_drift.py` analyze. Build it here, from the completed consensus run, or those lanes cannot run at all.

```
python scripts/template_observations.py RUN_DIR/controls/consensus.json \
--out RUN_DIR/templates/observed_templates.json \
--exceptions RUN_DIR/templates/observed_template_exceptions.json
```

A template is one document family, not one page. Labels are the union of what a family's documents showed, each carrying the number of documents it appeared in, so capture variance stays visible instead of producing one template per page. Labels are copied exactly as printed and ordered deterministically rather than in layout order, so drift detection sees an added or removed label and not a reordering. Nothing here is a mapping.

This is also why template drift is computed after consensus rather than in Phase 2: its input does not exist until extraction has run. A changed vendor layout is therefore detected after that layout has been extracted.

6. Run business controls

```
python scripts/address_normalize.py RUN_DIR/controls/validated.json \
--out RUN_DIR/controls/address_normalized.json \
--exceptions RUN_DIR/review/address_exceptions.json
python scripts/entity_resolve.py RUN_DIR/controls/address_normalized.json \
--out RUN_DIR/controls/parties.json \
--log RUN_DIR/controls/entity_merges.json
python scripts/attribution.py RUN_DIR/controls/proofed.json \
--reference REFERENCE_EXPORT \
--out RUN_DIR/controls/attributed.json \
--register RUN_DIR/review/unattributable.json
python scripts/completeness.py RUN_DIR/controls/attributed.json \
--gl GL_EXPORT --payments PAYMENT_EXPORT \
--out RUN_DIR/controls/completeness.json
python scripts/sampling.py RUN_DIR/controls/attributed.json \
--materiality MATERIALITY --out RUN_DIR/controls/sample_plan.json
```

Name the client with `entity_resolve.py --client-name`, as the pages print it, or the client becomes its own largest counterparty. Give `--party-decisions` a person's decisions about named pairs – `same_party`, `branch_of` (both accounts kept, the branch linked to its parent), `different_parties` and `not_a_party` – under the authorization the file names. A decided pair leaves pending adjudication with its score kept, and a decision the resolver cannot apply is reported with its reason rather than guessed at.

Recovering a printed column the schema had no field for

A schema that names no home for a printed column does not make an engine skip it: the value goes into whichever field looks closest, both vendors pick the same one, and it arrives accepted. Before proposing a re-extraction to capture such a column, reconcile from what the run already retains. Every provider response

is kept, and the independent non-LLM extractor retains the whole page rather than the fields the schema asked for.

```
python scripts/specifier_recover.py RUN_DIR/controls/validated.json \
--extractor-raw RUN_DIR/providers/raw/docai \
--out RUN_DIR/controls/records_with_specifiers.json \
--exceptions RUN_DIR/review/specifier_recovery_exceptions.json
```

The join is a printed key both sides already carry – these forms group lines under a header reading COMPANY / ACK# / P.O.# / P.O. DATE / SPECIFIER, so the specifier printed beside a purchase-order number belongs to every line quoting it. A recovered reading is one extractor with no model behind it, so it is written unaccepted as a single reading and never carries the corroboration status that pairs a model with an extractor. A line already stating a specifier keeps its own, and a page printing a specifier no line claims yields an exception rather than a guess.

Every lane reading the extractor’s pages indexes them through `run_io.retained_responses`, which takes the response that read the page. A request that came back without text is retried and both files are kept, and on the commission run 222 of 716 pages had text only in the retry: indexing the first file per page left all 222 empty to brand recovery, to this lane and to the page review’s extractor evidence.

`brand_recover.py` recovers the manufacturer the same way, from the retained page rather than a new call. It reads the letterhead first – the name at the top wins when a letterhead names two, and a name of ten letters or more may be read one letter off – then gives a page naming no maker the one its opening shares with that maker’s other pages. A continuation page prints its maker’s columns and none of its letterhead, so a page still unsettled takes the maker only whose settled pages print at least three of its words, or else the maker every one of its five most alike settled pages names. A page whose top names a company by its legal form, neither the client nor a known maker, stays an exception naming that company: Crestline Design’s funds-transfer advices read like Marlow/Bramwell’s bill payments, and a test over the pages the vocabulary already knows cannot see a maker it lacks.

After reviewing every certainty and MUS selection, provide a `sample_review_results_v1` artifact containing the plan’s `sample_plan_sha256` and exactly one explicit outcome per selected document. Missing, duplicate, foreign, stale, abstained, failed, or contradictory outcomes block the accuracy statement; an empty file is never a zero-error result.

Address processing is local unless Google Address Validation is explicitly enabled. Provider results add derived evidence and never replace the raw address. Entity merges are reversible. Attribution enumerates every unresolved in-scope dollar. Completeness blocks on missing or unmeasurable reconciliation inputs, out-of-tolerance periods, unresolved/ambiguous payments, sequence or calendar gaps, and undated documents.

Run `handwriting_review.py` only with independent HTR results. Printed and handwritten evidence remain separate. Handwritten financial readings require strict reconciliation and still become review-required amendments.

The optional Google handwriting lane uses Cloud Vision’s handwriting-capable `DOCUMENT_TEXT_DETECTION` feature on every retained page. Supply one or more separately produced extraction/profile artifacts containing normalized `handwriting_regions`; Google reads those regions but does not prove that its own reading is correct:

```
python scripts/google_handwriting_ocr.py \
  RUN_DIR/intake/ingestion_manifest.json \
  --regions RUN_DIR/extraction/primary_records.json \
  --out RUN_DIR/handwriting/google_htr.json \
  --adapter-out RUN_DIR/handwriting/google_htr_adapter.json \
  --exceptions RUN_DIR/handwriting/google_htr_exceptions.json \
  --raw-dir RUN_DIR/handwriting/google_htr_raw \
  --images-dir RUN_DIR/handwriting/google_htr_pages --enable
```

Every page receives one success/failure record. A page without a supplied region is recorded as OCR-complete but not declared handwriting-free. Freeze the raw response and rendered-page directories. Retry

provider failures in a new directory, then reconcile the resulting `google_htr.json` with genuinely separate HTR provider groups. The other inputs may be HTR annotation artifacts or retained visual-extraction record lists; each region/reading pair must share the same stable `region_id`:

```
python scripts/handwriting_review.py \
  RUN_DIR/handwriting/google_htr.json \
  RUN_DIR/extraction/openai_records.json OTHER_PROVIDER_HTR.json \
  --out RUN_DIR/handwriting/decisions.json \
  --exceptions RUN_DIR/handwriting/exceptions.json
```

Three independent groups must unanimously agree on handwritten numerics; two independent groups suffice for free text. Same-provider models and prompt roles count once. Signatures and financial amendments remain review work.

7. Build and reduce the review surface safely

Three options here are omissions rather than refinements, and each fails quietly. `--resolved-by` retains a document-type proposal the run already answered instead of asking it again; without it a real corpus re-asked 1,121 settled questions. `--classifications` gives grouping the document type the extraction consensus declined to claim – it writes "unknown" wherever its lanes could not agree, which was 691 of 718 documents on a real corpus, and without the classification artifact those become one undifferentiated `unclassified_document_family` group and the family is guessed from words in the reason string. The pack lane needs both, because it builds the same queue by its own path.

```
python scripts/final_review_queue.py RUN_DIR/review/*.json \
  RUN_DIR/controls/validated.json \
  RUN_DIR/controls/completeness.json \
  --resolved-by RUN_DIR/controls/classification_consensus.json \
  --out RUN_DIR/review/final_client_review.json
python scripts/review_grouping.py RUN_DIR/review/final_client_review.json \
  --consensus RUN_DIR/controls/consensus.json \
  --classifications RUN_DIR/controls/classification_consensus.json \
  --out RUN_DIR/review/grouping.json
python scripts/client_review_lane.py build RUN_DIR/review/*.json \
  --consensus RUN_DIR/controls/consensus.json \
  --classifications RUN_DIR/controls/classification_consensus.json \
  --resolved-by RUN_DIR/controls/classification_consensus.json \
  --manifest RUN_DIR/pages/ingestion_manifest.json \
  --out-dir RUN_DIR/review/client_pack_01
python scripts/operations.py review-export \
  RUN_DIR/review/final_client_review.json \
  --csv RUN_DIR/review/final_client_review.csv \
  --html RUN_DIR/review/final_client_review.html \
  --xlsx RUN_DIR/review/final_client_review.xlsx
```

The JSON queue is authoritative. CSV, HTML, and XLSX are reviewer aids. The card-level reviewer, cross-record pass, full-dataset agent, grouping, and safe consolidation are optional and disabled by default. They retain the original queue and may reduce only the client-facing presentation of explicitly eligible low-risk dependencies.

All review-reduction paths use the shared fail-closed protected-item taxonomy. Unknown reason codes are protected. Cross-record proof must come through the strict `cross_record_evidence_v1` envelope and one exact versioned producer handoff. `intake_evidence_producer_v1` binds the complete intake manifest, classification-exception artifact, retained source/page/text files, and operations manifest. The reducer reruns the retained deterministic text extraction and classifier and admits only rule-classified `pages[].document_type`. `canonical_evidence_producer_v1` binds the exact canonical export to its checksummed load plan, clear final-review artifact, source PDFs, and operations manifest; only `tables.document[].document_type` is eligible. Extra semantic fields, self-labelled tuples, arbitrary JSON, malformed entries, raw/provider update fields, suggestions, and unresolved or non-clear facts are reported as ineligible context.

Evidence preflight is local and precedes provider I/O. By default each envelope or required producer/source artifact is capped at 10,000,000 bytes and the supplied collection is capped at 10,000 declared entries. Positive overrides are available through `--max-evidence-artifact-bytes`, `--max-evidence-entries`, and their documented environment settings. A breach emits an explicit review exception without a provider call. The reducer first size-bounds and parses every envelope and totals all declared entries; no producer is loaded when the aggregate limit fails. Every producer, handoff, source PDF, page PDF, and retained text file is size-checked before its first hash, open, or read. Producer indexes are built once; authenticated source hashes, PDF validity, and page counts are cached so repeated entries read and open each retained source once. Even an empty collection reports an invalid producer once at `$.producer`; for a non-empty invalid producer, every declared entry is separately retained, with otherwise valid entries marked `producer_not_authenticated`.

A proposed update must target the reviewed field, use exact non-empty whitespace-free target and different source identities, pass finite non-Boolean `[0, 1]` confidence validation, and resolve one entry. Every failure remains review work with a machine reason. Eligible updates remain proposal-only and record the exact envelope, producer, and source provenance. Protected findings, including provider/schema, arithmetic, re-assembly, handwriting, identity, review-flag, missing-field, and unresolved-evidence items, cannot be batch cleared.

When authorized and calibrated, run `safe_review_consolidation.py` into a fresh directory. With grouping enabled it generates the small reproducible workbook, plain-language guide, decision pages, and ZIP. The package intentionally has no client appendix or drill-down; the internal JSON retains the exhaustive detail.

8. Process a returned client workbook

Preserve the issued workbook and the returned copy separately. Import them together:

```
python scripts/client_review_package.py import-decisions \
    RETURNED.xlsx --issued-workbook ISSUED.xlsx \
    --out RUN_DIR/review/client_decisions.json
```

The importer validates package structure, hidden safe-consolidation lineage, group completeness/order, fixed cells, controlled choices, protected decisions, editable columns, notes for client-supplied alternatives, ZIP safety limits, and both workbook hashes. Its output is proposal-only.

If the returned workbook is legacy or otherwise blocked by lineage, preserve the exact client choices and notes before troubleshooting the package:

```
python scripts/client_review_package.py preserve-responses \
    RETURNED.xlsx --issued-workbook ISSUED.xlsx \
    --out RUN_DIR/review/client_responses_preserved.json
```

This no-clobber, proposal-only artifact records every decision row, response count, workbook hashes, fixed-cell comparison, and any lineage exception. It does not alter either workbook and should accompany any reissued package.

When an authoritative client reference, GL, or payment export is missing, keep that control explicitly blocked. An operator may create a separate, hash-bound proposal package to prioritize follow-up, but it cannot turn the completeness gate clear:

```
python scripts/inferred_controls.py RUN_DIR/controls/consensus_or_proofed.json \
    --out-dir RUN_DIR/review/inferred_controls_proposal
```

For bounded post-return reference reasoning, first construct context from the preserved response artifact and a deterministic selection of retained records:

```
python scripts/client_review_context.py \
    RUN_DIR/review/client_responses_preserved.json \
    --records RUN_DIR/controls/consensus_or_proofed.json \
    --out RUN_DIR/review/client_review_context.json
```

If the client supplied comments outside the returned workbook, parse one or more general comment files into the same context contract, then include the result in source-native corpus forward and refinement calls:

```
python scripts/client_input_comments.py client_notes.json client_followup.csv \
  --out RUN_DIR/review/client_input_comments_context.json
python scripts/table_comprehension_corpus.py \
  RUN_DIR/intake/ingestion_manifest.json \
  --client-context RUN_DIR/review/client_input_comments_context.json \
  --out RUN_DIR/tables/corpus_with_client_context
```

JSON may be a list or an object containing `comments[]`; CSV/TSV requires a `client_comment`, `comment`, `comments`, `note`, or `text` column; TXT and Markdown preserve each non-empty line. Every comment and source field remains verbatim and hash-bound. It may guide terminology, priorities, and questions, but cannot supply document facts, independent consensus, authorization, or control clearance. The context hash is part of resumable corpus identity.

The context marks every client comment as reasoning-only and excludes it from independent consensus. The legacy one-pass discovery lane (`client_review_inference.py`) and the preferred iterative lane are both disabled unless explicitly invoked. The iterative lane uses a configured Primary LLM and independently configured Secondary LLM, retains raw responses and a record-level exception for any item that cannot fit its bounded packet, and accepts the full retained corpus only through `--records`:

```
python scripts/client_review_iterative.py \
  RUN_DIR/review/client_review_context.json \
  --primary-out RUN_DIR/review/iterative_primary.json \
  --buddy-out RUN_DIR/review/iterative_buddy.json \
  --exceptions RUN_DIR/review/iterative_exceptions.json \
  --raw-dir RUN_DIR/review/iterative_raw \
  --records RUN_DIR/controls/consensus_or_proofed.json --iterations 3 \
  --convergence-min-new-candidates 1
```

The lane adapts serialized packet boundaries and reserves 40,000 bytes for instructions, carried proposals, and buddy decisions; it retains every raw request and response, and writes exact `source_document_ids` into every exhausted batch exception. Retries are bounded; recovery uses a separate, no-clobber retry run for the named records, and any residual remains explicit review work. Its `confirmed` and `close_needs_review` states are still source-linked proposals only. They cannot establish GL/payment facts, approve attribution, clear completeness, or authorize a canonical fact. Build the deterministic JSON evidence graph from the selected, hash-bound artifacts when graph queries will help a reviewer; its `build`, `query`, `path`, `candidates`, `contradictions`, `schema`, and `schema-surface` commands preserve evidence and conflicts rather than resolve them. The bounded `schema` inventory reports observed field names, counts, examples, and claim provenance for proposal-only schema discovery. `schema-surface` reads an exact retained consensus artifact and separately reports non-consensus candidate fields, source pointers, and label-bound address/contact/representative signals. It retains all raw signals and provides only a bounded, source-cited signal summary to schema discovery; it never turns a candidate into a graph fact or canonical mapping. See [references/artifact-contracts.md](#) and [references/command-line-reference.md](#) for the exact artifact manifest and CLI contracts.

The extraction contract has a broad controlled core for commercial references, party/contact roles, payment/tax, sales, commission, and logistics fields. It also requires a source-labelled extension record for every other visible labeled value. This gives unfamiliar document families a retained, source-cited path into schema review instead of discarding an out-of-vocabulary field. The extension is proposal-only: a later mapping still needs evidence and approval.

The iterative primary/buddy provider, model, optional credential-variable override, timeout, retry, backoff, reasoning effort, and packet limits are all read from `.env` and recorded in the outputs. The code rejects a missing model, unsupported provider, or matching primary/buddy provider before provider I/O; safe code validation never silently selects a provider for a production run.

To propose next steps for retained reassembly, validation, or attribution exceptions, use the separate disabled-

by-default control-exception lane. It receives immutable records, named control findings, and—when client feedback is available—the preserved `client_review_context_v1` as reasoning-only context. It uses configured independent primary and buddy providers, preserves raw responses, and never changes the source records or clears a gate:

```
python scripts/client_review_exception_resolution.py \
  RUN_DIR/controls/consensus_or_proofed.json \
  --exceptions RUN_DIR/controls/arithmetic.json \
  --exceptions RUN_DIR/controls/validation_exceptions.json \
  --exceptions RUN_DIR/controls/attribution_register.json \
  --client-context RUN_DIR/review/client_review_context.json \
  --out RUN_DIR/review/exception_resolution_proposals.json \
  --exceptions-out RUN_DIR/review/exception_resolution_exceptions.json \
  --raw-dir RUN_DIR/review/exception_resolution_raw --enable
```

This lane uses the shared serialized-byte adaptive packet utility. It combines all supplied findings for one source record, reduces packet size before provider I/O, and records any individually oversized record or positive batch-cap remainder explicitly. It rejects duplicate record IDs, accounts for findings whose source record is absent, and gives the buddy reviewer the same findings, source records, and reasoning-only client context as the primary reviewer. A primary proposal is retained as rejected if its document or related-document ID falls outside the packet or its evidence quote is not present in the compact source packet. Review and apply only eligible append-only proposals, then rerun reassembly, arithmetic, validation, attribution, completeness, and final review. Inferred document evidence cannot clear authoritative GL or payment/remittance reconciliation. Every input finding has an exception ID, and every retained proposal names the source exception IDs it addresses so later overlays can be reconciled without guessing from batch membership. Provider/schema failures, individually oversized work, and cap-deferred work also retain the exact original findings, control kinds, and exception IDs. Passing that companion exception artifact to a new no-clobber recovery therefore reconstructs the original control packet rather than substituting a generic provider-failure finding or guessing from document membership.

AI simulated-client review comments

`scripts/ai_simulated_client_review.py` is intentionally separate from both the client workbook import and the client-review reduction lane. It accepts a final review queue, retained source-evidence artifacts, and optional JSON context artifacts, then asks one configured LLM to produce a source-cited draft recommendation for every item in each card. The default configuration selects OpenAI `gpt-5.6-luna`, but the provider and model are environment-configured and retained in the output; there is no hard-coded provider decision in the lane.

The package contains authoritative comment JSON, a separate exception artifact, one raw response per card, and derived CSV/XLSX reviewer views. Every generated comment begins “**AI client reviewed (simulated; not client authorization)**”. The provider packet stores each source record once and gives every review item its own source-reference list. The reducer validates each evidence quote within one scalar value of a source record assigned to that item, retains the exact artifact-scoped reference and JSON pointer, preserves invalid responses separately, and creates a comment or explicit exception for each final-review item. It automatically keeps all protected findings in human review even when the model recommends a simulated acceptance.

Context clipping is measured in UTF-8 bytes. The raw directory contains an atomic checkpoint bound to the queue, evidence hashes, and complete resolved non-secret configuration. `--resume` continues only unfinished cards when all bindings still match. After completion, `--retry-exceptions` creates a separate no-clobber recovery overlay for retained provider/schema, size, cap, missing-source, or explicit post-run evidence-audit cards. It never overwrites or silently merges the frozen base.

This is decision support, not delegated authority: it does not modify the issued or returned workbook, impersonate a client, authorize an amendment, clear a deterministic control, satisfy unavailable GL/payment evidence, or allow canonical/CRM staging. A real client or authorized operator must make a separate

decision, authorise a specific append-only change, and rerun the affected controls.

If an operator intentionally stops a later pass after earlier passes complete, do not rerun the earlier provider calls or overwrite their raw evidence. Write new final artifact paths and use `--finalize-existing` with the same retained raw directory and the intended completed-pass count. This recovery path makes no provider call; it reconstructs only valid retained primary/buddy pairs and records malformed or unpaired packets as explicit exceptions.

Each completed pass records its count and identifiers of new material relationships: a relationship is material only when it is source-backed, new to the run, and independently buddy-confirmed. The lane stops early when that count is below `--convergence-min-new-candidates` (default 1), or otherwise at its configured maximum. This is a cost and quality control, not approval.

```
python scripts/evidence_graph.py schema RUN_DIR/review/evidence_graph.json \
  --sample-limit 10 --out RUN_DIR/review/observed_schema_inventory.json

python scripts/evidence_graph.py schema-surface RUN_DIR/review/evidence_graph.json \
  --consensus RUN_DIR/controls/consensus.json --sample-limit 10 \
  --out RUN_DIR/review/schema_surface_inventory.json

python scripts/evidence_graph.py schema-recovery-scope RUN_DIR/review/evidence_graph.json \
  --consensus RUN_DIR/controls/consensus.json --topic address \
  --out RUN_DIR/review/address_recovery_scope.json

python scripts/schema_discovery.py discover RUN_DIR/review/source_templates.json \
  --graph-schema-inventory RUN_DIR/review/observed_schema_inventory.json \
  --enable --out RUN_DIR/review/schema_discovery.json \
  --exceptions RUN_DIR/review/schema_discovery_exceptions.json \
  --handoff-out RUN_DIR/review/schema_discovery_handoff.json \
  --raw-dir RUN_DIR/review/schema_discovery_raw
```

The schema inventory is validated, 65,536-byte-bounded corpus context for the LLM; it can inform vocabulary exploration but cannot map a source label by itself. Every mapping still requires the supplied source-template/page evidence and remains a client-review proposal. The Primary LLM makes the schema proposal; the Secondary LLM independently classifies each mapping against the same source packet. A confirmed, source-evidenced proposal at the configured primary confidence floor may be labelled `inferred_high_confidence`, but remains client-review-required and cannot alter a registry or canonical fact. The discovery output and provider handoff retain both inventory hashes and provider metadata for replay.

For the next cross-packet refinement, run a fresh proposal overlay after the iterative outputs and graph exist. The command has two independent, bounded worklists: it discovers links for documents with no in-packet buddy candidate, and verifies each already-resolved in-packet proposal against documents sharing a source-visible business identifier. The Primary and Secondary LLMs independently classify the existing proposal as `confirmed`, `close_needs_review`, `conflict`, or `unsupported`; neither result rewrites the prior proposal or authorizes a fact. Its primary and buddy provider/model/credential/timeout/retry settings are separate `.env` values, so a retry can deliberately use a different independent provider pairing while retaining the effective pairing in the artifact. The default allows two convergence-controlled passes. After the initial pass, the system builds a follow-up only from unresolved documents in newly exposed, source-visible neighborhoods of that pass's material candidates. It stops when a completed pass adds fewer than the configured minimum of new material relationships; it never expands from a provider assertion alone. Use separate verification and convergence caps to control cost:

```
python scripts/client_review_cross_packet.py RUN_DIR/review/client_review_context.json \
  --records RUN_DIR/controls/consensus_or_proofed.json \
  --primary RUN_DIR/review/iterative_primary.json \
  --buddy RUN_DIR/review/iterative_buddy.json \
  --graph RUN_DIR/review/evidence_graph.json \
  --max-verification-neighborhoods 100 \
  --max-iterations 2 --convergence-min-new-candidates 1 \
```

```
--out RUN_DIR/review/cross_packet_proposals.json \
--exceptions RUN_DIR/review/cross_packet_exceptions.json \
--raw-dir RUN_DIR/review/cross_packet_raw
```

To retry a failed run, pass its retained exception artifact with `--retry-exceptions`. The selector includes only discovery or verification neighborhoods that reached a provider failure; cap-deferred and identifier-less coverage findings are not silently converted into retry work.

For a completed iterative or cross-packet pass, use this as a finalization boundary: freeze the completed pass, select every provider-retryable exception, write the retry results to a new output directory, and reconcile that overlay with the frozen pass. Generate final proposal, exception, graph, and reconciliation manifests only after the retry reconciliation. Data, cap, identifier, and unsupported exceptions remain explicit in the final package.

It records every selected neighborhood, source identifier, raw request/response, provider decision, exception document ID, per-pass worklist, material-yield count, and exact termination reason. Cap-deferred and identifier-less items are explicit coverage exceptions. The resulting candidates and verification records are an immutable overlay; rebuild the graph with the new artifact and retain unresolved, close, conflict, and unsupported items for review.

Create an explicit change catalog for each actionable decision. Each entry names the append-only change type and target, proposed value, retained evidence, affected review items, documents, fields, and optional exact affected amount. Compile it against the safe consolidation artifact:

```
python scripts/client_decision_compile.py compile \
  RUN_DIR/review/client_decisions.json \
  RUN_DIR/review/safe_review_consolidation.json \
  RUN_DIR/review/client_change_catalog.json \
  --out RUN_DIR/review/compiled_change_plan.json \
  --authorization-template RUN_DIR/review/operator_authorization_template.json
```

The compiler rejects incomplete imports, workbook/consolidation hash mismatch, missing groups, protected batch changes, client-choice/catalog mismatch, and any claimed item/document impact outside the selected group. It shows the exact patch surface and minimum rerun lanes and writes a content hash. The authorization template defaults every patch to `defer`. After reviewing the plan, an authorized operator supplies their ID, an offset-bearing timestamp, and one `authorize` or `defer` choice per patch, then binds that decision to the unchanged plan:

```
python scripts/client_decision_compile.py authorize \
  RUN_DIR/review/compiled_change_plan.json \
  RUN_DIR/review/operator_authorization_completed.json \
  --out RUN_DIR/review/authorized_change_plan.json
```

This still does not change production facts. Apply only authorized patches to the appropriate append-only mapping, amendment, template, document-type, allocation, or source-quality configuration. For authorized template rules, `template_drift.py registry-update` can create the next no-clobber registry snapshot. Then:

1. Rerun only the affected source-quality, classification, mapping, or extraction lane.
2. Rerun consensus and applicable table reconciliation.
3. Rerun arithmetic and deterministic validation.
4. Rerun applicable address, entity, attribution, payment, and completeness controls.
5. Rebuild the exhaustive final review queue.
6. Regenerate safe consolidation and the client package.
7. Confirm the final gate is clear before claiming application or approval.

9. Produce analytics and approved-fact outputs

Descriptive analytics may be built earlier, but financial analytics remain blocked unless their required completeness, attribution, arithmetic, and review gates are clear.

```
python scripts/analytics.py RUN_DIR/controls/validated.json \
--review RUN_DIR/review/final_client_review.json \
--parties RUN_DIR/controls/parties.json \
--completeness RUN_DIR/controls/completeness.json \
--out RUN_DIR/controls/analytics.json
```

Only after the final review gate clears:

```
python scripts/canonical_deploy.py \
--out RUN_DIR/canonical/deployment_plan.json
python scripts/canonical_load.py RUN_DIR/canonical/canonical_export.json \
--out RUN_DIR/canonical/load_plan.json
python scripts/csv_api_staging.py \
RUN_DIR/canonical/canonical_export.json RUN_DIR/canonical/load_plan.json \
--out RUN_DIR/canonical/no_send_staging
python scripts/crm_import_package.py \
RUN_DIR/canonical/canonical_export.json RUN_DIR/canonical/load_plan.json \
--out RUN_DIR/canonical/crm_import
python scripts/retrieval_store.py build \
RUN_DIR/canonical/canonical_export.json \
--out RUN_DIR/canonical/business_retrieval.sqlite
./scripts/run_retrieval_mcp.sh \
RUN_DIR/canonical/business_retrieval.sqlite
```

Activate approved-fact MCP access for a completed production run

MCP activation is a Phase 6 operation. It must not be used as an alternate view of unfinished extraction, provider responses, review proposals, or an empty canonical directory. Before registration, require selected-lane coverage, `run_workspace.py` audit, a clear exhaustive final-review queue, every applicable business gate, the approved canonical export and read exception artifact, the exact load plan, and a non-empty retrieval snapshot built without `--allow-empty`.

Use `skills/mcp-api-operations/SKILL.md` as the governing setup, deployment, connection, acceptance, use, monitoring, rebuild, and shutdown procedure. Before deploying a client snapshot, establish repository code readiness with the fictional no-clobber acceptance:

```
PYTHON_BIN=.venv/bin/python bash scripts/run_mcp_production_acceptance.sh \
/tmp/mcp-production-acceptance
```

Require its versioned summary to report six passing gates, twelve local tools, fourteen remote tools, seven standard reports, and retained source/artifact hashes. This is code-integration evidence, not acceptance of the production proxy/TLS, identity provider, selected client workspace, operations, or exact approved client snapshot.

For Claude Code on the workstation holding the completed run, use private local scope and absolute paths:

```
claude mcp add --transport stdio --scope local business-document-retrieval -- \
bash /ABSOLUTE/REPOSITORY/scripts/run_retrieval_mcp.sh \
/ABSOLUTE/RUN/canonical/business_retrieval.sqlite
claude mcp get business-document-retrieval
```

Use `/mcp` to verify the connection. First call `get_crm_capabilities` and `get_crm_export_summary`; record the snapshot's source-export checksum, batch, registry, table counts, and integrity status. Then inspect representative schemas and exercise search, exact lookup, an account card, currency-partitioned sales, applicable reports, and bounded pagination. Rebuild a new snapshot after any authorized fact, mapping, amendment, or affected-control rerun; never mutate the served SQLite file in place.

The local stdio server works with Claude Code, Claude Desktop, and compatible local desktop clients. The included authenticated HTTPS listener is a REST API, not a remote MCP connector. `retrieval_remote_mcp.py` is the implemented stateless Streamable HTTP MCP for authorized Claude or ChatGPT workspace/API surfaces; plan, workspace-publication, and mobile availability must be verified during live

client acceptance rather than inferred from protocol compatibility. It requires a public HTTPS deployment plus RFC 7662 OAuth introspection and enforces token activity, expiry, audience, subject, exact tenant, allowed role, and `crm:read/crm:export` scope before business access. It also applies rate, payload, and optional Origin limits and writes content-free audit events. Public hosting, identity-provider registration, user-consent/deprovisioning policy, workspace review, monitoring, retention, and incident controls remain client-accepted deployment work. See `references/mcp-production-integration.md` for the full activation checklist, client options, and acceptance sequence. Use the exact public endpoint, including `/mcp`, as the OAuth resource and token audience. The server exposes both root and path-specific protected-resource metadata and serves existing 2025 initialize clients plus the 2026-07-28 stateless protocol with required routing-header/body agreement and modern result markers. Claude remote connectors are reached from Anthropic's cloud even when the user is in Desktop or mobile; add the custom web connector in the Claude account/organization first, then use that connected tool from iOS or Android.

The local retrieval layer exposes citation-preserving approved facts only. The SQLite build also embeds every validated canonical row, its idempotency key, its row hash, a per-table count/checksum manifest, the source-export and registry hashes, and a cross-table FTS5 index joined back to the authoritative row store. The database is built in a temporary sibling, integrity-checked, and published without replacing an existing file. Read paths recheck the applicable table count/checksum manifest before table query, exact lookup, or indexed CRM search, and verify each returned record hash. Local MCP, remote MCP, and HTTPS call one shared read-only service rather than implementing separate query logic. All provide document search/get, CRM capability and schema discovery, indexed cross-table record search, exact lookup of any canonical row, bounded field filters/projection/sorting/pagination, checksummed table export, joined account/address/contact cards, multidimensional sales analysis, and seven deterministic standard reports.

The remote MCP adds `create_crm_export` and `get_crm_export_job`. It creates a bounded CSV/XLSX export from one approved table or deterministic report, or a ZIP package for the approved staging/common-import artifacts when the deployment is configured with the exact canonical export and strict load plan. Each job records the snapshot/source hash, requester hash, exact arguments, optional review-link metadata, file checksum, size, and expiry, and returns a random short-lived signed download URL. Status is owner-bound; the token is stored only as a hash; the artifact is rechecked before download; expired, mismatched, missing, or modified files fail closed. Paginated jobs refuse a mid-export snapshot change, malformed manifests fail as controlled errors, and failed creation removes its private partial job directory. Formula-leading spreadsheet values are neutralized. These artifacts are read-only views and never authorize or perform a target-CRM write.

For a client question where the canonical key is unknown, start with `search_crm_records`; when it is known, use `get_crm_record`. Use `get_account_card` to pull the party, role/name history, addresses, contacts, invoice/payment activity, and source documents together. Use `query_crm_records` for governed one-table slicing with `eq`, `ne`, `contains`, `starts_with`, `in`, `gte`, `lte`, or `is_null` conditions. Use `analyze_sales` to group by up to three of company/customer, vendor, region, selling location, month, quarter, year, currency, ACK, or job while filtering dates, dimensions, and invoice-value bounds. Currency is always an effective grouping dimension, and account-card monetary summaries are separated by currency, so unlike currencies are never combined. The built-in report names are `account_directory`, `invoice_register`, `receivables_aging`, `sales_by_customer`, `sales_by_selling_location`, `attribution_coverage`, and `shipment_performance`. Each result is source-export-bound and checksummed; an aggregate retains its contributing source document IDs. Receivables aging buckets the current approved balance snapshot against the required `as_of` date, excludes later invoices, and is not a historical balance reconstruction. Attribution coverage counts invoice-target, non-failure attribution only and reports each currency separately.

The load planner requires `auto_accepted`, `sampled_verified`, or `exception_resolved` on every batch-managed row and rejects any other supplied status on every canonical row, including vocabulary and association/history rows. It then validates document source file/page/hash provenance and each table's declared parent chain. The no-send staging adapter recomputes that strict plan for the exact export, writes declared schema headings even for empty tables, records file/row/column and canonical-row checksums, and places authoritative compact canonical JSON plus a row SHA-256 beside the formula-safe display columns. Its v3 receiver envelope uses URL-safe batch endpoint templates and defines stage/reject/reconcile/rollback

contracts. Verification reconstructs every exact canonical row and requires the ordered file and receiver-operation topology before atomically publishing the package. It rejects legacy open-review fields. Open-review records, raw responses, and unapproved suggestions have no planning, staging, or factual retrieval override. PostgreSQL execution, HTTPS binding, target delivery, and public remote-MCP deployment require their explicit flags and separate authorization.

Export the field grain when canonical admits nothing

`canonical_export.py` admits only review-clear documents and has no override. On a corpus where every document carries an open exception that is zero documents, while most extracted fields are already accepted, and those readings have no way out of the artifacts. `field_extract_export.py` writes every field of every document exactly once with the evidence that accepted it or the reason it is withheld. It is not a canonical load and clears no control.

```
python scripts/field_extract_export.py RUN_DIR/controls/records_with_specifiers.json \
--manifest RUN_DIR/pages/ingestion_manifest.json \
--parties RUN_DIR/controls/parties.json \
--out RUN_DIR/delivery/crm_input.json \
--csv RUN_DIR/delivery/crm_input_fields.csv \
--lines-csv RUN_DIR/delivery/crm_input_lines.csv \
--documents-csv RUN_DIR/delivery/crm_input_documents.csv
```

Point it at the run’s **current** record artifact, not its consensus artifact: acceptances are appended after consensus, and reading one step too early understates them badly. Pass `--parties` or `resolved_party` is empty on every row, which reads as a corpus whose names resolve to nothing.

The pivoted grains carry columns a CRM needs and a pivoted cell cannot show for itself:

Column	What it says
<column>__amount, <column>__iso, <column>__month, <column>__currency, <column>__percent	The typed value beside the reading the page gave. <code>__month</code> holds a month the page names without a day (YYYY-MM), and <code>__percent</code> a rate only in the unit the page proves. The reading is never overwritten.
unreadable_in_source	The page never rendered this cell – <code>####</code> , scientific notation, a cut-off value. The typed money column carries <code>-99999</code> , which no real amount can be, so a load that ignores the flag still fails loudly. Not written into a date companion, which cannot hold it.
commission_arithmetic	<code>reconciles</code> , <code>does_not_reconcile</code> , or empty where a term is missing. A failing line is carried unchanged: most failures are the document contradicting itself, and correcting a printed number would invent evidence.
duplicate_of_line, restates_a_total, repeats_an_earlier_block, repeats_a_job_and_amount_above, amount_without_line_identity	Rows that are not a line of this document, and which a page total must exclude. The document’s own total rows – <code>P.O. Total:</code> , or a bare <code>Total :</code> – extract as lines, correctly, and summing them counts that money twice.
differs_from_the_same_line_elsewhere, missing_where_the_same_line_carries_it	The columns where this row disagrees with the same line on other statements, or is empty where they carry a value. Review, never exclusion: nothing is changed and the majority is the corpus repeating itself, not a vendor agreeing. It is the only evidence this corpus offers for fields no arithmetic can test.

Column	What it says
<code>repeats_a_line_in</code>	A real line of this document that also appears on another statement. A decision for the importer – filter it to load projects, keep it to load statements – and never an exclusion when totalling a single page.
<code>text_truncated_in_source, completion_proposed</code>	The page cut this reading off mid-word. A completion is proposed only where the corpus offers exactly one.
<code><column>__resolved_party, <column>__party_key</code>	The party master’s name and key for a party column. Join on the key: a name is not an identity.
<code>fields_naming_the_client</code>	The company fields on this row that name the client rather than a counterparty.
<code>party_fields_not_a_party</code>	Each company field whose reading the party master refused, and why: a number with no letter, a heading or placeholder, a total’s label or a signed adjustment, a street address; or a name a party decision removed.
<code>person_fields_not_one_person</code>	Each person field that carries a number, a single word, a role or report word, the client, a brand, or several people.
<code><column>__email</code>	The reading, only when it is an email address.
<code>row_confidence, unconfirmed_fields</code>	The weakest cell in the row, and which cells made it so.

Check the export against the pages, then fill and validate it

An export is only as good as its agreement with the pages. These lanes check, complete and validate it, in this order. None of them is canonical or clears a control.

1. `page_review.py` verifies the export against the source page rather than against another model. `--packet` emits one page’s image with the rows the export claims for it; the lane reads the reviewers’ returns from `--reviews`, names lost money (printed, held nowhere) and invented money (counted, never printed) mechanically, and `--proposals` writes what the reviews propose, graded by the independent evidence beside each correction.
2. `page_review_amend.py` applies those proposals as append-only amendments under the operator authorization `--authorization` names, at the `--evidence` grades chosen – by default every grade independent of the reviewer. It refuses a correction that would break a line’s own base x rate = commission, and holds a figure moved between lines until both halves are proved. Rebuild the export from the amended records.
3. `party_locations.py` is optional and off unless enabled, with `--enabled` or in the environment. It asks Google Places where each dealer and brand trades, one request per party within `--max-requests`, and accepts a candidate only when it accounts for every word of the party’s name. `--countries` limits where an accepted business may trade, `--skip-known` skips parties a public-source file already covers, and `--from-lookups` decides again from a previous run’s retained candidates, sending nothing.
4. `augment_export.py` fills empty cells beside the reading from the run’s own evidence, and `--external` adds operator-compiled public-source facts, accepted Places locations among them. Each inference sits in `<column>__inferred` with its method and evidence. With `--parties` it also writes the accounts (`--out-accounts`, joined on `<field>__party_key`, never from the client’s fields, a job or a refused reading) and the contacts (`--out-contacts`, one person’s spellings joined on evidence, a company only where an email naming the person proves it) a CRM loads beside the grains. With `--product-rules` it reads each line’s product by its manufacturer’s own rule and `--out-products` writes one product per manufacturer and code; `--extractor-raw` checks each code against the row the line’s own amounts are printed on. An inferred party carries the master’s key in `<column>__inferred_party_key`, and `same_key_on_other_lines` fills a line’s customer or dealer from the same maker’s lines printing its

order, job, project or transaction number, where every such line names one party. It is scored first on held-out lines, and a field under 95% fills nothing.

5. `crm_input_validate.py` checks each `--grain` against the canonical `--schema`. Nothing is repaired: every finding is a mapping decision.

A page score compares a page's amounts as a set, so it cannot see a figure moved to the wrong line. Check each line's own arithmetic before and after any change to line-level money.

Prepare common CRM imports without writing to a CRM

`crm_import_package.py` is the optional Phase 6 bridge between the full canonical package and the file-import surfaces commonly used by CRM teams. It creates twelve dependency-ordered UTF-8 CSVs covering accounts, roles, addresses, contacts, products, sales locations, shipments, transactions, transaction lines, charges, payments, and payment applications. Every row retains its canonical source table, exact idempotency key, authoritative row JSON, and row SHA-256. The manifest binds the export and strict load plan, checksums every artifact, and accounts for all 28 canonical tables as mapped or explicitly outside the common profile. Zero common rows remain an explicit finding.

The package also includes `field_mapping_template.csv`, `crm_write_plan.json`, and `official_import_sources.json`. Suggestions for Salesforce, HubSpot, Dynamics/Dataverse, and Zoho are starting points only. Any target-specific receiver or importer belongs on the destination system side; this repository stops at the verified no-send package boundary. The target owner must export the actual tenant metadata, complete and approve required fields/types, unique external or alternate keys, relationships, owners, currencies, enumerations, and automation effects, then prove a small sandbox run is idempotent and reversible. Any production mutation requires separate authorization bound to the exact package hashes. An adapter must retain rejects, record target job and record IDs, capture update before-images, and reconcile attempted rows to created, updated, unchanged, and rejected rows for every object. It may not hard-delete. The complete target contract is `references/crm-write-readiness.md`.

10. Account for every lane before calling the run complete

```
python scripts/pipeline_plan.py RUN_DIR \
  --out RUN_DIR/analytics/pipeline_plan.json
python scripts/run_lane_coverage.py RUN_DIR --out RUN_DIR/lane_coverage.json
```

Rule 9 holds that a control which processed nothing has not passed. Nothing enforced the same rule one level up: a lane nobody invoked leaves no artifact, no exception, and no trace, so a run can reach delivery looking complete while whole controls were never run. A full trial reached CRM staging with fourteen lane families never invoked, and no command anywhere said so.

This report scans the run directory for the artifacts each lane leaves behind and names the lanes that left none. It reads retained files only, contacts no provider, and changes nothing. A lane may be deliberately skipped; stating that out loud is the point. Add `--require LANE` for each lane an engagement must include, and the command fails rather than reports.

The catalogue it checks against is itself verified by `scripts/agent_surface_check.py`, which fails when a command in the repository is neither a catalogued lane nor explicitly exempt. That check also runs every command documented in an operating surface through its own parser, so an instruction that would fail at the point of use fails the release gate first.

Methodology

Consensus and model reasoning

Independent extraction reduces correlated reading errors. Provider identity, model, page hash, input mode, raw response, retry history, and schema exceptions are retained in versioned, content-hashed handoffs. Consensus refuses duplicate provider groups or lanes and verifies retained raw evidence before voting. Self-

reported model confidence is useful for later calibration but does not count as evidence and is ignored by consensus and approval routing.

The same principle applies to table comprehension: profile, extraction, and audit prompts may improve reasoning, but they are not separate voters when they use the same model. Cross-record and full-dataset agents can expose patterns and contradictions; they cannot turn their own reasoning into approved facts.

Arithmetic, attribution, and completeness

Arithmetic proves that a captured document is internally coherent. It does not prove that the document belongs to the correct entity, that all documents were captured, or that the source was itself correct.

Attribution answers where an in-scope monetary amount belongs. Every unresolved dollar is retained in the unattributable register with reason and amount.

Completeness compares the captured population with external expectations: sequences, calendar patterns, GL totals, and payment/aging closure. Missing or ambiguous evidence blocks the gate; no variance is plugged.

Sampling and accuracy language

Sampling draws only from eligible, review-clear records with arithmetic status **proved** or **not_applicable**. Monetary-unit sampling reports a one-sided 95% upper bound on **overstatement**. It does not measure understatement or missing documents. Attribute samples and golden-set field/template measurements answer different questions and must be reported separately.

No production accuracy or reduced-review claim is valid until all enabled lanes are measured on a representative client-authorized golden set, including protected-field precision, false auto-accepts, arithmetic proof, completeness, cross-record usefulness, and expected review exceptions.

Use `golden_set_evaluate.py` to make that measurement reproducible. Golden truth and pipeline predictions use unique item IDs and exact values. The report shows precision, recall, usable prediction coverage, abstentions/failures, auto-accept precision, false auto-accepts, protected escapes, expected-review recall, review reduction, and arithmetic/completeness mismatches overall and by template fingerprint, document family, field, risk category, and provider. Optional per-item latency, cost, and review-minute telemetry is summed without affecting exact-value correctness.

```
python scripts/golden_set_evaluate.py \
  RUN_DIR/acceptance/golden_truth.json \
  RUN_DIR/acceptance/pipeline_predictions.json \
  --out RUN_DIR/acceptance/golden_set_evaluation.json \
  --exceptions RUN_DIR/acceptance/golden_set_exceptions.json
```

The defaults require 0.99 precision, 0.95 recall, complete usable prediction coverage, and zero false auto-accepts or protected escapes. Tune thresholds only through a recorded client acceptance design. Every failed check produces a blocking exception item, even when no individual row has a separate failure. A pass is evidence for client acceptance review; `automation_authorized` always remains false.

Mappings, policies, and amendments

LLM schema discovery, Places candidates, table mappings, allocation policies, client decisions, and adjudication results are proposals. A mapping is effective only when an exact source-template fingerprint and source label match a client-approved append-only registry rule. Allocation rules are scoped and effective-dated. Corrections create new amendments; they do not revise raw evidence or historical rules in place.

Run `template_drift.py analyze` before mapping reuse. It fingerprints normalized document family, exact ordered headers, and supplied source-layout sections, totals, table regions, and column boundaries. One exact approved fingerprint permits reuse; a changed, new, or ambiguously duplicated layout creates review work. Similarity candidates help the reviewer find a likely predecessor but never enable a mapping.

Monitoring, recovery, and troubleshooting

Live progress and spend

`run_monitor.py` reads retained response counts and non-secret service summaries for LLM providers, Document AI, Address Validation, and Places. It writes atomic JSON snapshots and estimated USD spend. Supply an accepted price file when contract pricing differs. Estimates must be reconciled with provider billing exports before reporting actual cost.

`llm_usage_report.py` records provider/model selection, tokens where available, cache behavior, attempts, rate-limit events, delays, skipped work, exceptions, and retained-response counts. It does not contact a provider.

Provider health, before and during a run

```
python scripts/run_sentinel.py preflight
python scripts/run_sentinel.py watch RUN_DIR --interval 30 --fail-fast
```

`preflight` resolves every enabled lane to its provider and credential variable and refuses when one has nothing behind it. For Google Document AI and Vision, an unset token variable is valid when Application Default Credentials can mint a fresh short-lived token; the sentinel verifies that local ADC path without retaining the token or calling a document provider. A missing long-lived key and unavailable ADC remain different faults with different fixes.

`watch` reads the retained raw responses beside a running run – the place where a failure states its own cause – and classifies each as a credential, configuration, or transient fault. `--fail-fast` exits non-zero on the first two so a supervising script can stop the run. A transient failure is what each lane’s own bounded retry already handles, and stopping for one would make the sentinel the outage; a credential or configuration fault will repeat on every remaining page, so every page allowed to run after it is money spent on a known failure.

Both are deterministic with respect to run artifacts: no document provider or model is called, tokens are never retained, and nothing but the optional report is written. The Google check may ask local `gcloud` to mint and immediately discard an ADC token. This exists because a full trial spent a corpus before anyone noticed Document AI returning HTTP 404 on all 18 pages. Every raw response said `document_ai_http_404` from the first page onward, and nothing read them until the run was over.

Failure recovery matrix

Failure	Required response
Source hash, page count, or ordering mismatch	Stop; quarantine the run; re-establish the immutable source and begin a new run
Existing output path	Preserve it; choose a new run-scoped destination
Provider timeout, 429, 5xx, or schema failure	Retain raw/error evidence; keep an explicit exception; resume only through the supported verified mechanism
Every page of one lane failing	Not review work – a broken lane. Read the cause with <code>run_sentinel.py watch</code> ; a credential or configuration fault will not improve with retries, and a lane that read nothing corroborates nothing
Oversized request/page/context	Lower scope or approved caps; split into bounded batches/slices; never truncate away evidence silently
Manifest hash/configuration changed	Start a new run; do not attach new work to the old state
Consensus disagreement or empty provider record	Keep the field/document in review; obtain independent evidence or client adjudication

Failure	Required response
Arithmetic failure	Inspect source and amendments; never change a value merely to force reconciliation
Missing GL/payment/reference input	Mark the applicable control incomplete or blocked; do not infer closure
Invalid returned workbook	Preserve both files; report exact validation errors; request a corrected return without editing it
Open review before canonical load	Stop; apply authorized proposals and rerun the required gates
Secret or client-data exposure	Follow SECURITY.md ; contain, rotate/revoke as needed, and do not continue ordinary processing
Control reports zero work done	Treat as a failure, not a pass. evidence_graph build and retrieval_store build refuse an empty result unless --allow-empty-graph / --allow-empty is passed; independent_table_reconcile.py emits independent_reconciliation_produced_no_comparison . Find the upstream artifact that produced nothing
Rejected GL, payment, or reference rows	The register in the artifact names each row and reason. Correct the export and rerun; a rejected row is never excluded from the baseline silently, and while any remains the completeness gate stays blocked
Implausible commission rate or ambiguous rate unit	allocation_policy.py routes it to critical review rather than classifying it. Confirm the source figure and its unit; do not raise the plausibility ceiling to make a figure classify
Address contradicts its own country	address_country_postal_mismatch is review work. Confirm the country against the source page rather than accepting either half
“Another lane invocation is already active”	The lane lock is a POSIX flock and cannot outlive its holder, so a live process exists. run_workspace.py run does not propagate signals to its child, so a killed wrapper leaves an orphaned adapter still calling the provider. Find it with ps -eo pid,ppid,command grep llm_adapter , kill by pattern, and confirm the lock released
“Corpus state belongs to a different model configuration”	The control working. Model and reasoning effort are recorded in the retained state and compared on resume. Restore the original configuration and resume, or start a fresh pass; never force a resume that would blend two reading regimes into one artifact
Table page shows conditional_buddy_check.available: false	The check was unavailable, not unnecessary: no independent evidence was supplied. The accompanying no_llm_reported_issue is a default written before the check runs. Pass --independent-evidence ; pages produced without it ran with that control disabled
Mapping-dependent control promotes nothing	Check template-fingerprint overlap before blaming the data. A fingerprint derives from the emitted label set, so re-extraction orphans every prior approval. Retain the result as a failure

Measurements that have been wrong

Operating decisions get made on numbers the operator computes from retained artifacts. Four recurring mistakes have each produced a confident wrong figure and driven real spend at the wrong remedy.

The retained record can contain the prompt. Some providers echo the rendered instructions back in the response body. Scanning a whole retained record for vocabulary will therefore match the *prompt*, scoring every page a hit – including pages carrying none of it. Extract only the model’s emitted output, per provider response shape, before counting.

Rates over different populations are not comparable. Lanes at different progress have processed different pages. A rate computed for one lane over 600 pages against another over 90 measures the corpus, not the engines, and the difference presents convincingly as a quality gap.

Reimplemented helpers drift from the code. `profile_headers()` yields header strings; passing the raw header objects to `template_fingerprint()` folds column coordinates into the hash and fragments one schema into dozens. Import the function the lane imports rather than reproducing its logic.

A filename is not a lane, and a key is not the key. `--out` is operator-chosen, two lanes may share a default name, and retained state may record completed work under a name other than the obvious one. Match on a distinctive artifact key and confirm the key exists before drawing a conclusion from its absence.

Reading a high exception rate

A high consensus exception rate is not automatically an extraction defect. Improving extraction can raise it: better reading yields more `source_labelled_field_proposals`, consensus claims nothing over those, and the denominator grows while the numerator does not fall.

Inspect the composition before choosing a remedy. Predominantly **single-engine** exceptions indicate coverage depth, which re-extraction can address. Predominantly **provider disagreement** means both engines read the field and differed – a question the client answers with a rule per document family, and one that no further extraction will resolve.

Common questions

Can two runs of the same model count as consensus? No. They may help audit a proposal but are correlated evidence.

Can a 0.99 confidence threshold approve a field? No. Self-reported confidence is retained as provenance and is not a term in any decision; protected categories and final approval remain unchanged.

Can an OpenRouter lane and an OpenAI lane be my two consensus engines? Only if the routed model is not an OpenAI model. Independence is resolved from the model vendor behind the router, so `openrouter` running `openai/gpt-5.6` and a direct OpenAI lane on the same model are refused as one reading. A routed model slug with no resolvable vendor prefix is refused outright.

A control reported no findings. Is that a pass? Only if it processed something. Check the coverage or ingestion counts in the artifact: a reconciliation with `corroborated: false`, a graph build refused for having no nodes, or a non-empty rejection register all mean the clean-looking result covers nothing.

Can the returned workbook update facts automatically? No. It produces validated proposals. An authorized append-only change and downstream rerun are required.

Can open-review rows be loaded for exploration? No. Canonical load planning, no-send target staging, and approved-fact retrieval all reject them without an override. Keep analytical experiments outside these approved-fact artifacts.

Why keep raw responses and multiple image variants? They make later reconciliation, model upgrades, and rule corrections auditable without replacing the source or repeating every provider call.

Security, privacy, and delivery

- Keep client data, raw provider responses, `.env`, caches, credentials, retrieval databases, and generated run outputs outside version control.
- Use separate restricted credentials for Address Validation, Places, Document AI OAuth, Vertex ADC, OpenAI, OpenRouter, and HTTPS retrieval.
- Record the provider, model, region, purpose, retention decision, and request cap before sending client evidence externally.
- Prefer local MCP for approved-fact retrieval. HTTPS requires explicit enablement, TLS, bearer authentication, approved bind/network controls, token rotation, logging, monitoring, incident response, and tenant isolation.
- Before delivery, regenerate each changed tracked Markdown/LaTeX/PDF triple, render every changed PDF to page images, and inspect margins, boundaries, tables, headers, wrapping, and legibility.

Extending the repository

Read `CONTRIBUTING.md`, `BRANCHING.md`, `SKILL.md`, and the relevant phase references first. Extend shared registries and control helpers instead of forking consensus, provider, review-protection, no-clobber, or safety logic.

For a new provider:

1. Implement the shared adapter/provider contract and isolated credential namespace.
2. Register lane/model resolution without adding hidden selection defaults.
3. Retain raw responses, page hashes, selected settings, usage, and failures.
4. Apply shared throttling, request preflight, retries, and circuit-breaker rules.
5. Add fixture, schema, exception, CLI, and full branch tests.
6. Update runtime configuration, command index, artifact contracts, phase reference, acceptance evidence, and release boundary.

For any interface change, update code, tests, `.env.example` if applicable, runtime/CLI references, artifact contracts, agent routing, changelog, and generated documents in the same change. `release_check.py` prevents unindexed environment settings or argparse commands from shipping.

Glossary

Term	Meaning
Amendment	Append-only proposed or authorized correction linked to original evidence
Canonical fact	Authorized, provenance-linked value that passed applicable controls and has no open review
Consensus	Agreement among genuinely independent extraction evidence under the deterministic field rules
Exception	Explicit unresolved, malformed, failed, or conflicting item that blocks or informs a gate
Golden set	Representative client-authorized corpus with accepted truth used for field/template evaluation
HTR	Handwritten text recognition; independent reconciliation is required for accepted readings
Proposal-only	May inform a decision but cannot mutate source evidence or approve a production fact

Term	Meaning
Protected item	Review finding that cannot be cleared by thresholds, dependency carry-forward, grouping, or batch decision
Registry rule	Versioned, client-approved exact mapping or policy with scope and effective history
Review card/group	Presentation aid that retains links to every underlying queue item; never an approval itself
Source-native row	Table reading expressed in the document's visible labels/layout before canonical mapping

Reference map

- `references/artifact-contracts.md` — exact artifact and client-package contracts.
- `references/runtime-configuration.md` — all environment tunables.
- `references/command-line-reference.md` — every executable, phase, and CLI control.
- `references/cli-help-catalogue.md` — every shipped positional input, option, default, choice, and `--help` description, generated from the live parsers.
- `references/workflow-gates.md` — phase entry/exit and final-gate rules.
- `references/operations.md` — manifests, state, privacy, adapters, and exports.
- `references/table-comprehension.md` — source-native table workflow.
- `references/automation-controls.md` — validation, HTR, and review automation.
- `references/allocation-policy.md` — effective-dated allocation decisions.
- `references/semantic-schema-discovery.md` — mapping and Places proposals.
- `references/canonical-deployment-retrieval.md` — post-review data boundary.
- `references/crm-write-readiness.md` — common import files, target discovery, mapping, sandbox, authorization, reconciliation, and rollback contract.
- `references/mcp-production-integration.md` — gated MCP activation, Claude registration, acceptance queries, and remote-client boundary.
- `references/image-intelligence-controls.md` — variants, detectors, and visual evidence controls.
- `references/qa-sampling.md` — sampling design and reporting limits.
- `references/attribution.md` — business-reference attribution method.
- `references/handwriting.md` — handwriting capture and reconciliation.